



COMSATS University Islamabad

Department of Computer Science

OS LAB Assignment 4

Instructor: Miss Faiqa Rashid

Course Title: CSC322-Operating Systems

Maximum Marks: Total = 15
Averaged to 10

Dated: 13/May/2026

Due Date: 17/May/2026

CLO-6: Compose Linux commands using Shell scripting.

Lab 12# Writing Shell Scripts using Conditional-Statements, and Loops

Task 1: (2 marks)

Write a shell script that asks the user to enter the marks for three subjects and calculates the GPA for each subject along with the CGPA.

Task 2: (2 marks)

Write a menu-driven shell script that gives four options A, B, C and Q to the user to select one of them. If user enters A then it displays the host-name and uptime, if user enters B then it gives information about disk and memory space, if user enter C then it gives information about home space utilization and if user enters Q then it quits the program.

Task 3: (2 marks)

Write a shell script that, given a filename as the argument will count vowels, blank spaces, characters, number of line and symbols.

Lab 13# Using Arrays, and Functions in Shell Scripts

Q1: (1 mark)

Write a shell script that contains two function fact() and is_prime(). fact() function finds the factorial of a number while is_prime() checks whether a number is prime or not.

Your script should take an integer as input from user and pass this number to fact() and is_prime() functions as argument.

Q2: (1 mark)

Write a shell script that contains a function play_game. This function generates a random number between 1 and 10; and keeps on asking the user to guess the number as long as user enters a number which is equal to the random number. In the end, the total number of attempts made by the user to enter correct guess is displayed.

Q3: (2 marks)

Write a function that finds the sum of digits in an integer

Q4: (2.5 marks)

Write a function that finds whether an integer is a palindrome or not

Q5: (2.5 marks)

Write a function that writes an integer in reverse (for example writes 123 as 321)